



HOSPITAL EMERGENCY ROOM Patient Analytics Project Report

Patient Volume | Admissions |
Demographics | Operational
Performance

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PREPARED AS A DATA ANALYTICS PORTFOLIO PROJECT

TOOLS USED: MICROSOFT EXCEL/CSV DATA PREPARATION AND MICROSOFT
POWER BI

This project analyses Hospital Emergency Room (ER) patient data to understand patient volume, admissions, demographics, waiting time and patient satisfaction. The analysis was presented through an interactive Power BI dashboard so that important operational indicators can be reviewed in a simple and business-friendly format. The report is based on the Hospital ER_Data dataset and the final Hospital Emergency Room – Patient Analytics dashboard. The dashboard covers patient records from April 2023 to October 2024.

PROJECT OVERVIEW

Business Objective

The main objective of this project is to convert raw emergency room patient data into useful information for management and operational review-

- Understand the overall patient volume and admission level.
- Monitor admission rate and compare admitted and non-admitted patients.
- Analyse patient demographics such as age, gender and race.
- Understand the distribution of patients across referral departments.
- Monitor average patient waiting time.
- Review available patient satisfaction information.
- Provide an interactive dashboard for quick filtering and decision support.

Dataset Description

The source dataset contains 9,216 patient records and 12 fields. It includes patient identification information, admission date, demographic information, referral department, admission status, satisfaction score, waiting time and the Patients CM field.

Field / Area	Description
Patient information	Patient ID, first initial and last name
Admission details	Admission date and admission flag
Demographics	Gender, age and race
Referral	Department referral information
Patient experience	Patient satisfaction score and waiting time
Additional field	Patients CM recorded as a binary 0/1 field

Data Preparation and Analysis Approach

THE DATA WAS REVIEWED AND PREPARED FOR ANALYSIS BEFORE BUILDING THE DASHBOARD. THE MAIN PREPARATION ACTIVITIES WERE FOCUSED ON DATE HANDLING, DEMOGRAPHIC GROUPING, REFERRAL CATEGORISATION AND KPI CREATION.

PATIENT ADMISSION DATE WAS USED FOR TIME-BASED ANALYSIS.

PATIENT AGES WERE GROUPED INTO 0-17, 18-29, 30-44, 45-59 AND 60+ CATEGORIES.

RECORDS WITHOUT A DEPARTMENT REFERRAL WERE PRESENTED AS 'NO REFERRAL' IN THE DASHBOARD.

ADMISSION STATUS WAS USED TO COMPARE ADMITTED AND NON-ADMITTED PATIENTS.

AVERAGE WAITING TIME AND AVERAGE SATISFACTION WERE USED AS OPERATIONAL AND EXPERIENCE INDICATORS.

A DATE-BASED FILTERING STRUCTURE WAS USED IN POWER BI FOR INTERACTIVE ANALYSIS.

Key Performance Indicators

The dashboard provides five primary KPI cards. These indicators give a quick view of the overall emergency room performance represented in the dataset.

KPI	Value	Interpretation
Total Patients	9,216	Total patient records available for analysis.
Total Admissions	4,612	Number of patients marked as admitted.
Admission Rate	50.04%	Approximately half of the patients were admitted.
Average Wait Time	35.26 minutes	Average recorded patient waiting time.
Average Satisfaction	4.998 / 10	Average satisfaction among records with a recorded score.

Dashboard Analysis and Key Findings

-Patient Volume

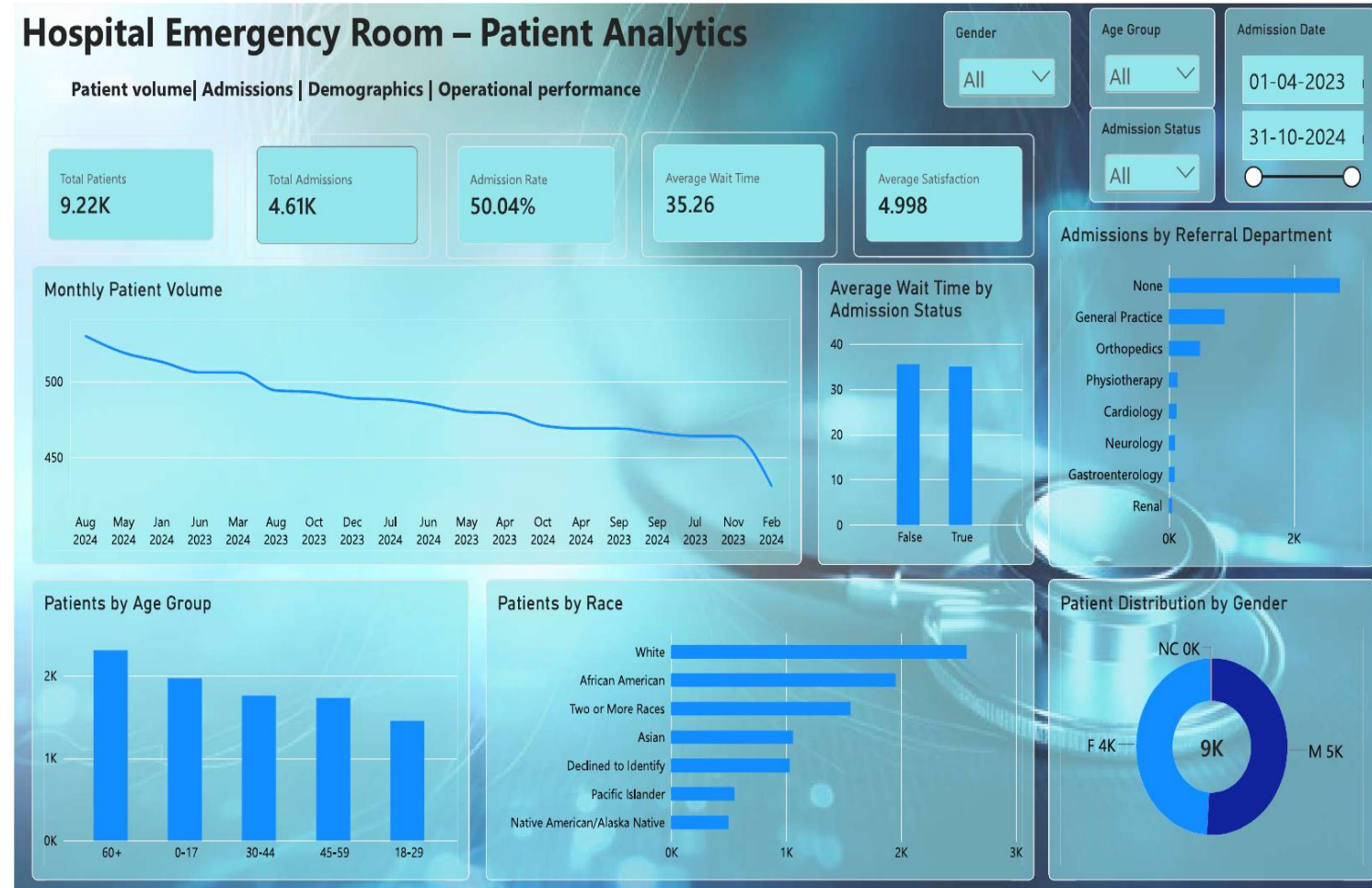
-Admissions

-Age Group Analysis

-Gender Analysis

-Race Distribution

-Referral Department



Operational Performance Analysis

7.1 Waiting Time

The overall average waiting time is 35.26 minutes. The dashboard also compares waiting time by admission status. The average waiting times are very similar for admitted and non-admitted patients, indicating that admission status alone does not show a major difference in the average waiting time in this dataset.

Admission Status	Average Wait Time
Not Admitted	35.55 minutes
Admitted	34.97 minutes
Overall	35.26 minutes

7.2 Patient Satisfaction

The average satisfaction score is 4.998 out of 10 among records where a satisfaction score is available. The dataset contains 2,517 recorded satisfaction scores out of 9,216 patient records, giving a response coverage of approximately 27.3%.

This is an important data-quality consideration. The average satisfaction score should be interpreted as the average of recorded responses and not as a score representing every patient.

Admission Status	Average Satisfaction
Not Admitted	4.91 / 10
Admitted	5.08 / 10

7.3 Satisfaction by Admission Status

Admitted patients have a slightly higher average recorded satisfaction score than non-admitted patients. The difference is relatively small and should be interpreted carefully because satisfaction responses are available for only a portion of the total patient population.



Dashboard Design

Key Business Insights

High overall patient volume: The dataset contains 9,216 emergency room patient records, providing a strong base for operational analysis.

Balanced admission outcome: The admission rate is 50.04%, with 4,612 admitted and 4,604 non-admitted patients.

Older patients form the largest age group: Patients aged 60+ account for 2,307 records and represent the largest age category.

Gender distribution is balanced: Male and female patient counts are close, with only a small difference between the two groups.

No-referral cases are significant: 5,400 records fall under No Referral, making it the largest referral category and an important area for operational review.

Waiting time is around 35 minutes: The overall average wait time is 35.26 minutes, with admitted and non-admitted patients showing similar averages.

Satisfaction data has limited coverage: Only about 27.3% of patient records have a satisfaction score, so satisfaction findings should be treated as response-based insights.

Patients by Age Group

2K

1K

0K

- Monitor waiting time regularly by admission status, age group and referral category to identify areas where operational improvement may be possible.
- Review the high number of No Referral records and confirm whether these cases represent genuine non-referral patients or missing referral information.
- Improve collection of patient satisfaction scores so that the organisation can obtain broader feedback from patients.
- Use the dashboard filters during regular operational reviews to compare patient activity across different dates and patient segments.
- Track patient volume and admission rate periodically to identify changes in emergency room demand.
- Use demographic analysis as a supporting operational indicator while avoiding assumptions about patient needs that are not directly present in the dataset.

60+

0-17

30-44

45-59

18-29

Recommendations

Conclusion

The Hospital Emergency Room – Patient Analytics project demonstrates how raw patient-level data can be converted into a clear and useful business dashboard. The analysis provides visibility into patient volume, admissions, demographics, referral patterns, waiting time and available satisfaction feedback.

The Power BI dashboard gives management a quick way to monitor the key indicators and apply filters for different patient segments and dates. The project also highlights the importance of data quality, especially when satisfaction and referral information are not available for every patient.

Overall, this project demonstrates practical skills in data preparation, data analysis, Power BI dashboard development, KPI reporting and business insight generation.

Project Deliverables

- Hospital ER patient dataset.
- Power BI Hospital Emergency Room – Patient Analytics dashboard.
- Project report documenting the analysis, findings and recommendations.

End of Report